Ask the Model: Counterfactual Cue Reliance Predicts Worst-Group Failure and Targets Repair

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Abstract

Fine-tuned models can rely on shortcut cues that correlate with labels during training but mislead on some inputs. We ask whether a finished model's behavior reveals how much it relies on such a cue. Given a suspected cue, we measure counterfactual cue reliance, Δ_{cue}, by editing the cue and measuring the change in the model's decision score. This requires no group labels, training data, or knowledge of the training recipe. We test 486 fine-tuned language models on CivilComments and Bias-in-Bios while varying the training cue–label correlation. Across 180 models in 10 main task–model settings, higher Δ_{cue} predicts lower worst-group accuracy. In a separate high-seed analysis, Δ_{cue} predicts at held-out correlation levels, where linear extrapolation from the assigned correlation performs worse than predicting the mean. Capability and placebo-edit controls show that neither model quality nor generic edit sensitivity explains the result. Δ_{cue} also provides signal for prioritizing concept-erasure repair. Exact control shows that most of the predictive signal comes from differences across correlation levels. Within a fixed recipe, the relationship across random seeds is small and not robust.

1 Introduction

Recent work in mechanistic interpretability has argued for a more pragmatic, action-oriented standard, where methods are evaluated by whether they support concrete empirical tasks and interventions (Nanda et al., 2025; Orgad et al., 2026). One natural test is whether a measurement taken from a finished model can predict a downstream failure that matters and help decide what to do about it.

Fine-tuned models provide a useful setting for this question because they can rely on shortcut cues. These are features that correlate with the label during training but become misleading on some inputs (Geirhos et al., 2020). For example, if identity terms co-occur with toxic labels during training, a toxicity classifier may learn to read an identity mention itself as evidence of toxicity. It then does worse on non-toxic comments that mention an identity (Borkan et al., 2019; Koh et al., 2021). Our question is whether the finished model itself can tell us how much it relies on such a shortcut.

We ask the model directly (Fig. 1). Given held-out inputs containing a suspected cue, we counterfactually edit the cue and measure the resulting change in the model's decision score. A larger average shift in the cue-consistent direction indicates greater reliance. For a classifier with decision score $s(x)$ and edited input $\tilde{x}$, we define counterfactual cue reliance as

$$\Delta_{\text{cue}} \;=\; \mathbb{E}_x[\, s(x) - s(\tilde{x}) \,]\,. \tag{1}$$

Once the cue and edit are specified, Δ_{cue} requires no group labels, no access to training data, and no knowledge of the training recipe. Counterfactual edits and model-reliance measurements are both well established (Garg et al., 2019; Prabhakaran et al., 2019; Kiritchenko and Mohammad, 2018; Fisher et al., 2019). We test whether this simple measurement, taken on its own from a finished model, predicts meaningful downstream failure, transfers to unseen training-correlation levels, and gives useful information for repair.

To test this, we train 486 decoder LMs on two shortcut-prone tasks while deliberately varying the training cue–label correlation $\rho \in \{0.55, \ldots, 0.97\}$ and random seed. The tasks are CivilComments toxicity classification with an identity-term cue (Borkan et al., 2019; Koh et al., 2021) and Bias-in-Bios profession classification with a gendered-word cue (De-Arteaga et al., 2019). Because ρ is experimentally assigned, we

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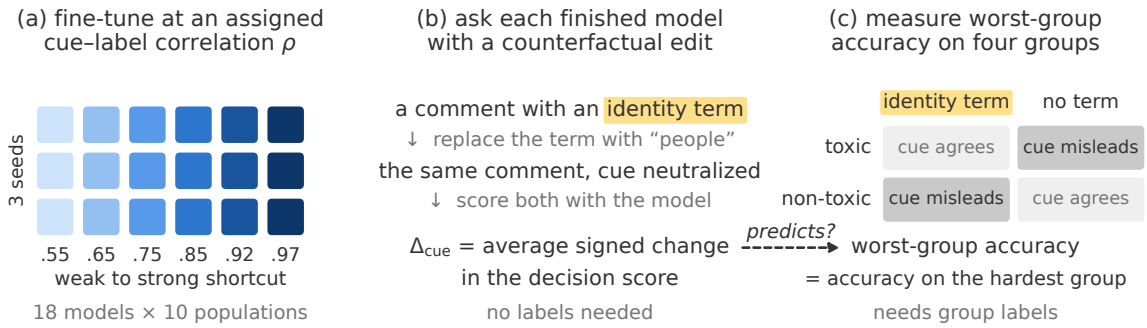


Figure 1: **Overview.** (a) Each of the 10 main populations has 18 LMs fine-tuned at six assigned cue–label correlations ρ. (b) Δ_{cue} is the average signed change in the decision score after a counterfactual cue edit. (c) Worst-group accuracy is the outcome it should predict. Panels (b) and (c) show CivilComments, and Figure 2 reports the test.

can directly compare what the finished model reveals with simple extrapolation from the assigned training correlation.

Contributions. We make three main findings:

- **Cue reliance predicts shortcut failure.** Across 180 models in 10 task–model settings, higher Δ_{cue} predicts lower worst-group accuracy. The prediction also holds at held-out training-correlation levels in a separate high-seed analysis. There, linear extrapolation from the assigned correlation performs worse than predicting the mean. Neither general model quality nor generic edit sensitivity explains the result (§4–5).
- **Cue reliance provides signal for repair.** We compare five post-hoc repairs under one protocol. LEACE improves mean worst-group accuracy in all six settings tested, and hard directional ablation can be harmful. Δ_{cue} also provides signal for choosing which models to repair with LEACE (§6).
- **Most of the predictive signal is across correlation levels.** Treating the discrete assigned correlation as a linear control initially made Δ_{cue} appear predictive beyond the training correlation. Exact control removes that apparent effect. Among models trained with the same recipe but different random seeds, the remaining relationship is small and statistically fragile (§7).

These results validate Δ_{cue} as a simple per-model audit. In the controlled populations we study, measuring what a finished model uses can tell us more about shortcut failure than extrapolating from the assigned training correlation.

2 Related Work

Our question connects work on shortcut measurement, cross-model prediction, and robust repair.

Measuring reliance. Counterfactual edits are widely used to test whether predictions depend on sensitive or spurious features (Garg et al., 2019; Prabhakaran et al., 2019; Veitch et al., 2021). Li et al. (2026a) identify token-level shortcuts from a converged model and mitigate them at deployment time, while Ravuru and Srivastava (2026) discover and causally verify text shortcuts for annotation-free debiasing. Related work also studies cue reliance through hidden-state bias directions and probe-based representation entanglement (Gupta et al., 2026; Singh, 2026), and its mitigation via consistency training (Chua et al., 2025; Imran et al., 2026). We ask a different question, namely whether a reliance score read off a finished model predicts worst-group accuracy across independently trained models and at held-out assigned cue–label correlations. The question is open because model-side bias measurements do not necessarily predict downstream bias (Goldfarb-Tarrant et al., 2021; Cao et al., 2022).

Predicting robustness across models. Identically trained models can generalize differently (McCoy et al., 2020; D'Amour et al., 2022), and prior work asks whether model-side measurements predict generalization across model populations (Jiang et al., 2020; Li et al., 2026b). Such comparisons can be driven by the design factors used to generate the population (Dziugaite et al., 2020), while in-distribution accuracy can itself predict OOD accuracy (Miller et al., 2021). Other work predicts per-example OOD behavior from internal causal features (Huang et al., 2025). We

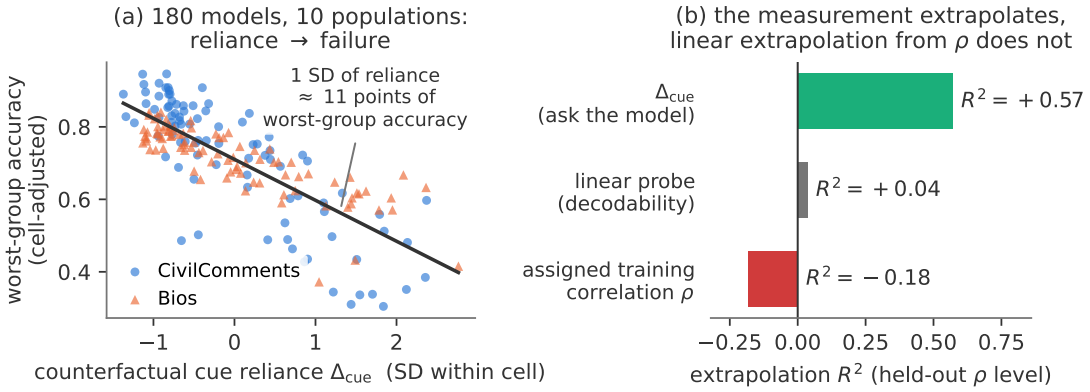


Figure 2: **Measured cue reliance predicts worst-group failure across training-correlation levels.** (a) β=−0.113 per within-cell SD of Δ_{cue}, partial R^2=0.63 beyond population identity. This pooled relationship includes the across-level effect. Section 7 isolates the within-level estimate. (b) Each predictor is fit on two ρ levels and predicts models at the held-out third. Δ_{cue} transfers, and linear extrapolation from ρ does worse than predicting the mean.

assign the cue–label correlation directly and test prediction at held-out levels of that assignment.

Repair. Worst-group robustness and post-hoc repairs are well studied (Sagawa et al., 2020; Liu et al., 2021; Kirichenko et al., 2023; Qiu et al., 2023; Belrose et al., 2023). Model-side measurements can guide selective debiasing, steering, and predictions of fine-tuning biases (Kuzmin et al., 2025; Billa, 2026; Lovering et al., 2021), while localization does not reliably identify editing layers (Hase et al., 2023). We work at the model level and ask whether a pre-repair score can rank independently trained models by how much they gain from the same repair. Concurrent work also studies repair-aware model selection using group-annotated validation performance (Wang et al., 2026). Our reliance score needs no group labels.

3 Methods

The design compares two ways of predicting failure. One measures the finished model. The other extrapolates from its assigned training correlation. That comparison needs a population in which we assign the shortcut signal instead of inferring it.

Populations. Each population *cell* is one task–model pair. Within each cell, we vary the assigned cue–label correlation $\rho \in \{0.55, 0.65, 0.75, 0.85, 0.92, 0.97\}$ and random seed. This creates models trained on the same task but exposed to shortcut signals of different known strengths. The main analysis has 10 cells from two tasks crossed with five model configurations: Qwen2.5-0.5B/1.5B, Pythia-410m/1.4b, and OLMo-2-1B. Each cell contains six correlation levels and three seeds, giving 18 models per cell and 180 models in total. A separate high-seed arm uses three cells (Qwen-civil, Qwen-bios, and Pythia-civil) at $\rho \in \{0.65, 0.85, 0.97\}$ with 15 seeds per stratum, giving 135 models. Throughout the paper, cell names without an explicit scale use the smallest model in that family. Thus, Qwen refers to Qwen2.5-0.5B and Pythia refers to Pythia-410m. Larger scales are named explicitly. Training sets contain 4,000 label-balanced examples and realize the assigned ρ exactly. Our main models use LoRA. Additional full-fine-tuning and instruction-tuned-base arms test whether the result depends on that choice. Across the main and auxiliary study arms, we train 486 models in total. Full training details are in Appendix B.1.

Tasks and edits. CivilComments uses 32 unambiguous identity terms as the cue (Appendix B.3), following the identity-term tradition of Dixon et al. (2018). The edit replaces each identity term with "people". Bios uses gendered words, and the edit swaps gender, applied only in the well-defined male→female direction. The two edits are different operations that measure the same target quantity (Eq. 1), and we verify in §5 that what they measure is cue-specific. Appendix B.2 audits both edits (coverage, minimality, and the reason the bios swap is used in one direction only).

Measurement. Each input is scored with a fixed prompt. For CivilComments it is `Comment: {text}`, a line break, `Question: Is this`

`comment toxic?`, a line break, and `Answer:`. For Bios the first two lines are `Bio: {text}` and `Question: Is this person a professor or a physician?`. The decision score $s(x)$ is the logit of the answer token `A` minus the logit of `B` at the last position. `A` stands for toxic or professor and `B` for non-toxic or physician. Each token has a leading space and is looked up in the model's own tokenizer. Δ_{cue} averages $s(x) - s(\tilde{x})$ over 300 cue-containing texts drawn from the same held-out pool as the evaluation groups. These texts never appear in training. They may partially overlap the evaluation sample, and no labels are used in computing Δ_{cue}.

Outcome. We evaluate worst-group accuracy over the four (label $\times$ cue-present) groups. These groups separate cases where the cue agrees with the label from those where it is misleading. For Bios, every pooled bio contains gendered words and cue-present means the bio is male, so the four groups are profession $\times$ gender. Worst-group accuracy uses all four groups, while the edit is applied only in the male$\rightarrow$female direction. We use a non-overlapping, deduplicated held-out split of 796–800 texts per group and sample 400 per group for each evaluation. Our grouping is coarser than the WILDS 16-cell definition (Koh et al., 2021), so the absolute numbers are not comparable to that leaderboard.

Statistical protocol. Because ρ is discrete and experimentally assigned, we control for it with indicator variables rather than treating it as linear. Section 7 shows why this matters. Population cells enter as fixed effects. The within-recipe estimator was pre-specified as the equal-weighted mean of the within-stratum slopes. Section 7 reports its robustness checks, and Appendix E.3 gives the additional instrumental-variable analysis.

4 Δ_{cue} Predicts Worst-Group Failure

Within-population prediction. In every population we train, models that react more to the cue edit do worse on the groups where the cue is misleading. The effect is large. When a model's measured reliance is one standard deviation higher, its worst-group accuracy is about eleven points lower ($\beta = -0.113$, $p = 3.1\times10^{-38}$, n=180, Fig. 2a). Knowing only which population a model comes from explains less than half of the variance ($R^2 = 0.46$). Adding Δ_{cue} raises this to 0.80, a partial R^2 of 0.63. The relationship also holds in every cell on its own (Fig. 3), so it is not an artifact of pooling.

Out-of-regime extrapolation. Within a designed population, both reliance and failure are consequences of the assigned ρ. A sharper test is to predict a correlation level the predictor has never seen. We fit the predictor on two ρ levels and predict models from the held-out third (high-seed arm, n=135, cell fixed effects). Δ_{cue} transfers to the held-out level, while simple extrapolation from the assigned correlation does not (Fig. 2b, Table 3). Extrapolating ρ itself performs worse than predicting the mean (R^2=-0.18). The reason is that ρ's effect on worst-group accuracy is strongly nonlinear (§7), so a line fitted on two levels misplaces the third. A nearest-level step extrapolation, the natural alternative with two observed levels, also performs worse than predicting the mean (R^2=-0.14). Because Δ_{cue} is measured after training, it can reflect how strongly the finished model came to rely on the cue. This result is mainly about placing models across correlation levels. Δ_{cue} does not consistently rank individual models within a held-out level (per-fold R^2 of -0.85, $+0.69$, and $+0.19$). We study fixed-recipe ranking directly in §7.

Robustness to the fine-tuning setup. The relationship remains similar under full-parameter fine-tuning and under an instruction-tuned base ($r = -0.905$ LoRA/base, -0.897 full FT, -0.907 instruct, n=18 each).

5 Validity Controls

The prediction would be uninformative if Δ_{cue} merely restated model competence or generic sensitivity to edits. We test both alternatives directly. Within our controlled populations, neither accounts for the result.

Capability control. In-distribution competence predicts robustness across models (Miller et al., 2021). In our own attempt to extend this study to public toxicity classifiers, capability explained nearly all of the variation (Spearman $+0.98$, see Limitations). Overall accuracy is not a clean capability control within our population because it already includes the worst group. We therefore build a control from non-overlapping example sets. We

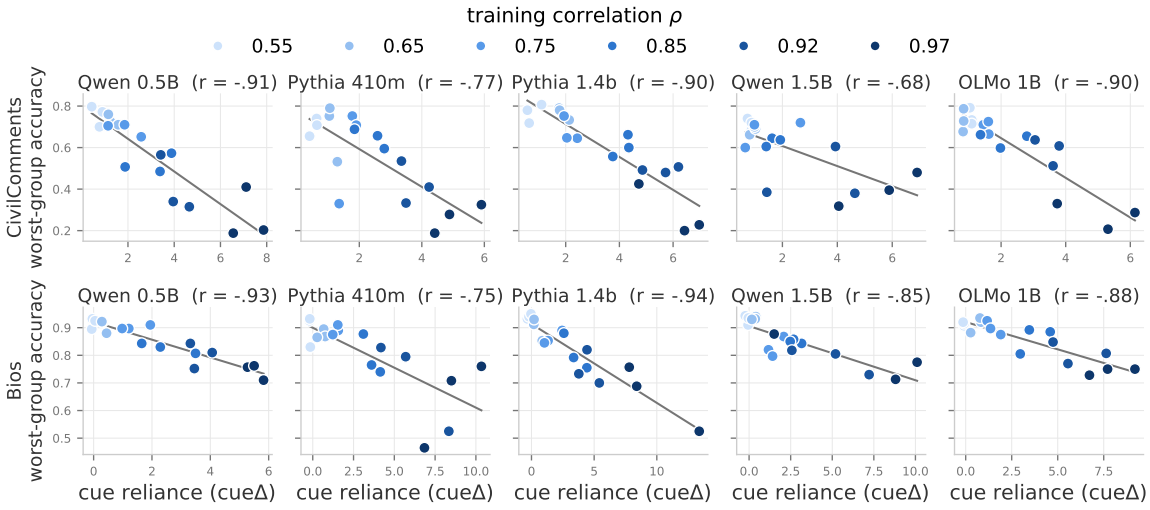


Figure 3: **The relationship holds in every population cell.** Each panel is one cell, and each point is one fine-tuned model shaded by its assigned ρ (per-cell r from -0.94 to -0.68, all $p < 0.002$). Because the points span the ρ levels, these correlations include the across-level effect (§7).

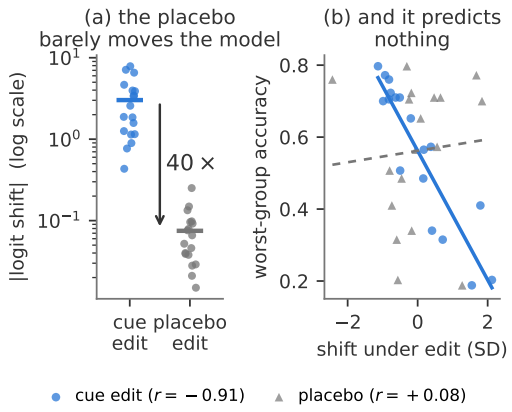


Figure 4: **Placebo edits barely move the model and predict nothing.** The same edit applied to 32 frequency-matched non-cue words shifts the decision score $40\times$ less (a). Across the same 18 models, the placebo shift does not track worst-group accuracy (r=+0.08), while Δ_{cue} does (r=−0.905) (b).

predict accuracy on the groups where the cue disagrees with the label from Δ_{cue} while controlling for accuracy on the groups where it agrees. On the 108 bios models that log all four groups, Δ_{cue} still strongly predicts accuracy on the harmed groups after this control ($\beta = -0.076$, $p = 8.0\times10^{-21}$), adding partial $R^2 = +0.25$. The capability control itself adds no detectable predictive value ($p = 0.77$).

Placebo-edit control. We repeat the same procedure using 32 frequency-matched non-cue words. These placebo edits move the model about

cell	ablate	prosp.	LEACE	DFR	AFR
Qwen-civil	+.067	+.056	+.064	+.180	+.002
Pythia-civil	−.262	−.208	+.111	+.186	+.048
OLMo-civil	−.185	−.169	+.079	+.160	+.020
Qwen-bios	+.043	+.035	+.028	+.079	+.003
Pythia-bios	+.039	+.059	+.060	+.128	+.051
OLMo-bios	+.049	+.052	+.038	+.084	+.024

Table 1: **LEACE improves mean worst-group accuracy in every cell, while hard ablation can be strongly harmful.** Mean change in worst-group accuracy (n=18/cell) for five post-hoc repairs: fixed-depth directional ablation, ablation at a prospectively selected layer, LEACE erasure, DFR-style group-balanced head refit, and AFR-style reweighting. Only DFR uses group labels.

$40\times$ less than the actual cue edits and do not predict worst-group accuracy (r=+0.08 versus r=−0.905 for Δ_{cue}, Fig. 4). So the measurement responds to the cue itself. Editing text in general does not produce it.

Reliable measurement. Split-half reliability of Δ_{cue} across models is 0.993 (Spearman–Brown 0.996, within-model SE 0.12 vs. between-model SD 2.32). The small within-recipe effects below are therefore unlikely to be explained by measurement noise alone.

6 Repair: What Works, and for Whom

A useful audit should predict failure and also say what to do about it.

Five repairs, one protocol. Table 1 compares five repairs, and three patterns stand out. First, LEACE improves mean worst-group accuracy in every cell. Second, hard directional ablation is very harmful in the civil cells of two of three families, and it gets worse with scale (-0.385 at Pythia-1.4b, -0.344 at Qwen-1.5B). This matches known failure modes of subspace interventions, namely damage to task-relevant features (Kumar et al., 2022) and off-manifold artifacts (Makelov et al., 2024). Third, DFR is the strongest repair everywhere, but it needs group labels. Our variant refits on training data, since the held-out split is our only evaluation. This may overstate its performance relative to Kirichenko et al. (2023). AFR, which uses class labels but no group labels, is near zero. LEACE's worst-group gains do not come with large overall-accuracy losses. Post-LEACE overall accuracy stays within 0.03 of label-dependent DFR's in every cell. The cells where ablation is harmful also show large overall-accuracy drops (0.63–0.69 vs. 0.77–0.79 under LEACE in the same cells, Appendix D.1).

Targeting: whom does erasure help? In practice, a practitioner with many fine-tuned models and a limited repair budget must decide which models to treat. Δ_{cue} provides signal for this decision. It predicts the gain from LEACE under exact ρ control in three of four specifications (Table 2, Fig. 5). The strictest specification falls to p=0.066, and some of the signal overlaps with baseline headroom. We therefore read this as useful targeting signal. It does not establish a definitive fixed-recipe ranking.

Allocation under a repair budget. We next ask whether this signal is useful for selecting models in practice. Within each high-seed cell, models are ranked before repair. At a budget of 15 out of 45 models, selection by Δ_{cue} gives a mean gain of $+0.200$, close to the oracle gain of $+0.205$ and above the $+0.091$ from random selection. Selection by observed failure or assigned ρ performs similarly at this budget. At smaller budgets, however, ρ cannot rank models within the same correlation level, while Δ_{cue} still can (Appendix D.2).

7 What Exact Control of ρ Reveals

A pitfall of linear control. Our own first analysis, mirroring common practice, entered ρ linearly and made Δ_{cue} appear strongly predictive

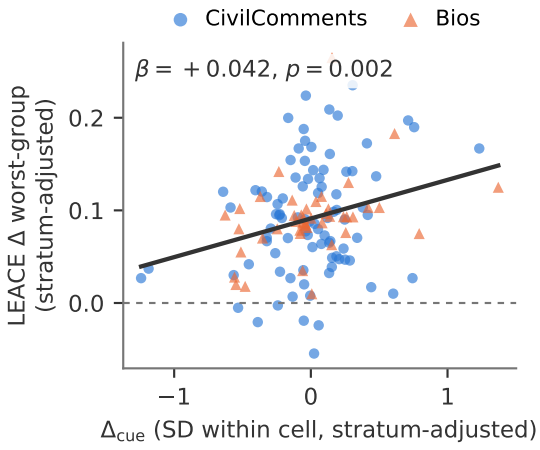


Figure 5: **Cue reliance provides signal for erasure gains.** Partial-regression view of the first row of Table 2, showing LEACE worst-group gain against Δ_{cue} with both adjusted for cell×ρ stratum (n=135). In this specification, higher Δ_{cue} is associated with larger gain.

Δ_{cue} standardized	baseline ctrl	β	p
within cell	no	+.0417	.0018
within cell	yes	+.0337	.0060
within stratum	no	+.0125	.0134
within stratum	yes	+.0086	.0657

Table 2: **Δ_{cue} provides repair-targeting signal in three of four specifications.** Post-LEACE worst-group gain regressed on Δ_{cue}. All rows include cell×ρ (stratum) fixed effects, n=135.

beyond the assigned training correlation. But ρ's effect is strongly nonlinear ($F(4, 165)$=13.4, p=1.7×10^{-9}, Fig. 6a). Linear control therefore leaves residual ρ variance that any correlated measurement can absorb. This is the classic residual-confounding problem (Groenwold et al., 2013). Table 4 and Fig. 6b show the pattern. The coefficient is significant under no control and under linear control, but not under indicators. With a discrete assigned factor, saturated adjustment uses four additional degrees of freedom (Lin, 2013). A linearly misspecified control, by contrast, can bias the coefficient of interest more than omitting the control entirely (Goff, 2014). For this discrete assigned factor, we avoid a linear specification. The population-of-models literature contains both practices today. Jiang et al. (2020) condition exactly and compute rank correlations within fixed settings of every other design factor. More recent work in the same genre controls its design grid with partial correlations on residuals (Tan et al.,

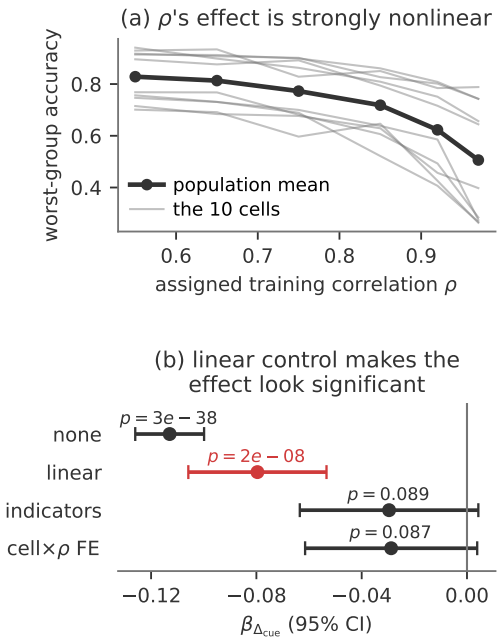


Figure 6: **Why linear control of ρ is unreliable.** (a) The effect of ρ on worst-group accuracy is flat at low ρ and steep at high ρ ($F(4, 165)$=13.4). (b) The Δ_{cue} coefficient under four controls for ρ. Its apparent significance under linear control disappears under exact control.

2024).

The within-recipe estimate. At fixed recipe, the signal is much weaker. The pre-specified equal-weighted estimator is negative and nominally significant (p=.012), but the result is not robust. A permutation test gives p=.061. Leave-one-cell-out checks range from p=.033 to p=.079, so the result is sensitive to which cell is removed. The pre-specified confirmation half gives p=.141. We therefore do not treat random-seed ranking within a fixed recipe as established.

The direction is nevertheless fairly consistent across strata. A conservative equivalence analysis rules out only effects larger than about 14% of the between-correlation effect, so the data also do not support a tight equivalence claim. Fixed-recipe differences therefore carry a small and statistically fragile signal. Most of Δ_{cue}'s predictive power comes from placing models across training-correlation levels without observing the assigned correlation.

"Used" vs. "decodable," scoped. Across training-correlation levels, Δ_{cue} predicts worst-group accuracy better than a linear probe for the cue. At fixed recipe, neither provides a robust ranking signal, and the probe is near ceiling. The comparison therefore supports the across-correlation result but is uninformative for seed-level ranking.

8 Cue Discovery and Label Efficiency

We close with two practical properties of the audit, cue discovery and label efficiency.

Automatic cue discovery. So far the audit assumes that the suspected cue is known. On CivilComments, we test how much this requirement can be relaxed. We replace the oracle lexicon with the top-20 PMI tokens from each model's own training draw, using add-100 smoothing (Gururangan et al., 2018). The discovered cue sets overlap the oracle by only $1.4/20$ words on average, yet reliance measured with them still predicts worst-group accuracy ($r = -0.55/-0.60/-0.63$ across civil cells). So the result does not depend on the hand-written lexicon. Cue discovery does, however, use the model's training draw, which the audit itself does not.

Label efficiency. A second practical advantage is that Δ_{cue} does not require group-labeled evaluation data. Using only 25 unlabeled cue-containing texts, it matches or beats an estimate based on 10 group-labeled examples per group in five of six cells. Direct group labels remain more useful in CivilComments, while on two Bios cells Δ_{cue} remains competitive even with substantially larger labeled sets. Appendix E.1 gives the full comparison. Appendix E.2 reports a related negative result on using early-training Δ_{cue} as an early-warning signal.

9 Discussion

Studies of this kind often explain model performance using the design factors that generated the model population (Morris et al., 2019; Jiang et al., 2020). Our results offer a complementary view. At held-out training-correlation levels, measuring the finished model can be more informative than simple extrapolation from the assigned correlation. The training correlation still matters, but its effect on worst-group accuracy is strongly nonlinear. Simple extrapolation can therefore fail, while Δ_{cue} measures the reliance the trained model actually expresses. Combined with the repair results, this provides a concrete example of actionable interpretability (Orgad et al., 2026). The measure-

ment predicts a downstream failure and informs the choice of which models to repair. Exact control also clarifies the scope of the result. Most of Δ_{cue}'s predictive power is across correlation levels, while the relationship within a fixed recipe is small and not robust.

10 Conclusion

This paper asked whether counterfactual cue reliance can serve as an actionable audit of a finished model. In the controlled populations we trained, the answer is yes, within a scope we can state precisely. Higher Δ_{cue} predicts lower worst-group accuracy. In a separate high-seed analysis, the prediction carries over to held-out training-correlation levels, where simple extrapolation from the assigned correlation performs poorly. Δ_{cue} also gives a useful signal for choosing which models to repair with LEACE. Exact control sets the scope. Most of the predictive signal is across correlation levels, and differences across random seeds within a fixed recipe are small and statistically fragile. For a practitioner with a fine-tuned classifier and a suspected cue, the lesson is simple. In these settings, what the finished model does with the cue tells you more about shortcut failure than extrapolating from the correlation it was trained on.
Supplementary code is available at:
https://github.com/prakharg55/cue_reliance.

Limitations

Scope and generalization. We study two English binary-classification tasks, decoder LMs of 0.4–1.5B parameters, and lexical cues. Structural or distributed cues are outside our scope. The result also does not extend directly to heterogeneous public checkpoints. Across eight public toxicity classifiers, capability dominated and Δ_{cue} was not predictive ($r{=}+0.06$). Whether reliance can be separated from capability across uncontrolled models remains open. The audit is label-free, but it still assumes a candidate cue and a meaningful counterfactual edit. Whether the result holds for substantially larger instruction-tuned models is untested.

Prediction limits. At fixed ρ, the relationship across random seeds is small and not robustly established (permutation $p{=}.061$, confirmation-half $p{=}.141$). The out-of-regime analysis tests only linear and nearest-level extrapolation from two observed ρ levels. With only two observed levels per fold, any nonlinear extrapolation requires an assumed shape that the data cannot determine. Stronger theory-motivated recipe predictors remain an open comparison. Alternative perturbation or attribution summaries also remain untested. Our probe baseline measures cue decodability. It does not measure task–cue direction entanglement.

Repair limits. Our DFR implementation refits on training data and may therefore overstate its performance. Repair targeting misses significance in its strictest specification ($p{=}.066$), and part of its signal overlaps with baseline headroom. Mediation-style decompositions of the ρ effect were unstable at our sample size and are not reported.

Ethics Statement

Both datasets are standard, publicly released research corpora, namely CivilComments (CC0) and Bias-in-Bios. The identity lexicon deliberately excludes ambiguous terms (e.g., *white*, *black*) that are unreliable identity references. We do not infer demographic attributes of any person, and we perform no dialect inference. Toxic example texts appear only in aggregate statistics. The ρ-controlled populations intentionally train models with amplified spurious behavior. All such models exist only as measurement logs, and none is released. Δ_{cue} is an audit measurement. We caution against interpreting low Δ_{cue} alone as certification of fairness, especially since it did not separate from capability on uncontrolled public models.

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A Reproducibility and Verification

All numbers in this paper are produced by analysis scripts from raw training logs and re-verified mechanically. A checker traces every reported value, including the tables in this appendix, to regenerated script output, and the figures assert their plotted values against the same captured results. The pre-specified primary analysis of §7 was locked before the confirmation half of the high-seed data was analyzed. Trained weights were not retained, so re-analysis of the models themselves requires retraining from the configuration in Appendix B.1. The training code, the analysis code and the logs of all trained models are in the repository linked in the Conclusion.

B Experimental Detail

This section records the training configuration and the data-side audits behind §3.

Across study arms, the frozen $2\times2\times2$ grid contains 144 models, the three-family grid contains 108, and the high-seed arm adds 108 new models. The robustness arms contain 36 models from Qwen-civil under full fine-tuning and an instruction-tuned base. The per-epoch trajectories contain 72 models from Qwen-civil and Pythia-civil, and the placebo arm contains 18 models

from Qwen-civil. This gives 486 trained models in total. All runs completed.

B.1 Training Configuration

All models are decoder LMs fine-tuned as binary classifiers scored by the logit difference of the two answer tokens at the last position. LoRA: r=16, α=32, dropout 0.05, applied to all attention and MLP projections. Optimization: AdamW, lr 2×10^{-4}, 4 epochs, batch 32 (civil) and 24 (bios), n_{train}=4,000, max length 96 (civil) and 160 (bios) tokens, fp32. The full fine-tuning arm uses lr 2×10^{-5} with no adapter. The instruction-tuned arm uses the same recipe on an instruct base. Families: Qwen2.5 (0.5B, 1.5B), Pythia (410m, 1.4b), OLMo-2 (1B). Δ_{cue} is estimated on 300 held-out texts that contain the cue per model.

Base checkpoints are Qwen2.5-0.5B, Qwen2.5-1.5B, and Qwen2.5-0.5B-Instruct, Pythia-410m and Pythia-1.4b, and OLMo-2-0425-1B, all from the Hugging Face Hub under the `Qwen`, `EleutherAI`, and `allenai` organizations. Each run used a single NVIDIA A100 GPU. The 490 scheduler jobs behind the reported results, covering the 486 fine-tuning runs and the public-classifier evaluations, total 83 GPU-hours, with a median of 7.5 minutes per job. Software: PyTorch 2.9.0, PEFT 0.19.1, scikit-learn 1.6.1, and statsmodels 0.14.2. The Transformers version used at training time was not recorded.

B.2 Counterfactual Edit Audit

We audited the realized correlations of actual training draws, re-detecting cue presence from the text itself instead of from the sampling key. At ρ=0.55 the realized $P(\text{toxic} \mid \text{cue})$ and $P(\text{toxic} \mid \text{no cue})$ are 0.550 and 0.450, at ρ=0.97 they are 0.970 and 0.030, and label balance is 0.500 in both audited draws.

Civil (identity term → "people"): held-out group sizes 796–800 per group, 1,598 eval texts that contain the cue, 100.0% changed by the edit, mean 1.50 tokens replaced per text (median 1, max 10), mean text length 49 words, edit fraction 3.1% of tokens.

Bios (labels professor vs. physician, male→female lexicon swap): group sizes 800 per group, 1,600 male-cue eval texts, 99.9% changed, mean 3.14 tokens swapped (median 3, max 10), mean length 60 words, edit fraction 5.2%. The swap does not undo itself. Applying it twice changes 67.4% of texts, because his and him both map to her and the reverse map returns his. The measurement therefore uses only the well-defined male→female direction.

predictor	R^2	MAE
Δ_{cue} (measured on the model)	+0.571	0.127
ρ (assigned correlation)	−0.182	0.219
nearest-level step (from ρ)	−0.140	0.220
linear probe (decodability)	+0.037	0.186

Table 3: **The model-side measurement extrapolates better than the assigned correlation.** Leave-one-ρ-out evaluation, fitting on two correlation levels and predicting the held-out models at the third (cell fixed effects, n=135). The step baseline predicts each held-out model with its cell's nearest observed level.

B.3 Cue and Placebo Lexicons

Identity lexicon (32 terms): arab, asian, atheist, bisexual, buddhist, catholic, christian, christianity, christians, feminist, gay, gays, hindu, hispanic, homosexual, immigrant, islam, islamic, jew, jewish, jews, latina, latino, lesbian, lgbt, lgbtq, mexican, mormon, muslim, muslims, queer, transgender. Ambiguous terms (e.g., white, black, american) are deliberately excluded as unreliable identity references.

Placebo lexicon (32 words, matched to the identity lexicon's per-word document-frequency profile on the 16,000-text training pool, total document frequency 10,151 vs. 10,184, screened to exclude identity content): attacking, believes, chose, court, damn, damned, deal, doctrines, having, heard, idea, know, laws, long, maher, makes, meaningless, member, movie, needs, oregon, paid, perhaps, right, sanders, should, sure, take, think, want, west, years.

In the placebo arm, shifts are 17.8× smaller after adjustment for edit coverage and are weakly correlated with Δ_{cue} (r=−0.11). The placebo–worst-group correlation is +0.08 (p=0.75), compared with −0.905 (p=2.4×10^{-7}) for the cue edit.

C Statistical Detail for the Population Results

This section gives the exact values behind the estimates of §4 and §7.

C.1 Extrapolation and Ladder Estimates

Tables 3 and 4 give the exact values behind Fig. 2b and Fig. 6b, including the nearest-level step baseline quoted in §4.

control for ρ	$\beta_{\Delta_{\text{cue}}}$	p
none	$-.1129$	3.1×10^{-38}
linear	$-.0796$	1.6×10^{-8}
indicators	$-.0296$	.089
cell$\times\rho$ FE	$-.0288$	.087

Table 4: **Linear control overstates the apparent effect of Δ_{cue}.** The same regression under four controls for ρ (n=180, cell fixed effects throughout). The apparent significance after linear control disappears under exact control.

C.2 Per-Stratum Detail

Table 5 gives the nine within-stratum regressions behind the pre-specified estimate of §7, and Table 6 the same effect under all four scaling choices. The estimate is strongest when both variables are standardized within stratum (p=.015) and weakest under the mismatched scaling, z within cell with stratum fixed effects (p=.147). The equivalence bound in Appendix C.3 deliberately uses that widest interval.

The pre-specified equal-weighted estimator is -0.027 accuracy points per Δ_{cue} unit ($t(8)$=-3.26, p=.012). Across the nine strata, the pooled correlation is -0.224 (95% CI $[-0.394, -0.039]$), with $Q(8)$=4.5 and I^2=0%. Seven of nine slopes are negative, with both positive slopes at ρ=0.97. The spread of Δ_{cue} increases from about 0.26 at ρ=0.65 to 1.30 at ρ=0.97, causing pooled fixed-effects OLS to place 76% of its weight on the ρ=0.97 strata and 3% on ρ=0.65. Robustness checks give permutation p=.061 and leave-one-cell-out p=.038/.079/.033 when dropping Pythia-civil, Qwen-bios, and Qwen-civil, respectively. The confirmation half gives p=.141 on the pre-specified seeds 9–14.

For the probe comparison, probe partial effects are null in 7 of 8 frozen cells. Within fixed strata, probe SD is 0.0007–0.0156, compared with 0.24–1.70 for Δ_{cue}, explaining why the fixed-recipe probe comparison is largely ceiling-limited.

C.3 Equivalence Bounds

The bounded interval is the 90% CI of the pooled regression of worst-group accuracy on within-cell-standardized Δ_{cue} with stratum fixed effects, the widest and therefore most conservative of the four scalings in Table 6. Scale anchors: within-stratum SD of worst-group accuracy 0.0737, and the between-correlation effect (ρ 0.97 vs. 0.65) -0.3875 accuracy points. Following Braun et al. (2022) we report the TOST verdict across margins (Table 7) rather than a single chosen margin. Principled candidate margins span 0.007 (a tenth of the within-stratum SD) to 0.039 (a tenth of the between-correlation effect), and none of them supports an equivalence claim.

cell	ρ	SD(Δ_{cue})	SD(w)	b	r
Pythia-civil	.65	0.249	.085	$-.042$	$-.123$
Pythia-civil	.85	0.452	.105	$-.054$	$-.233$
Pythia-civil	.97	0.966	.087	$+.010$	$+.107$
Qwen-bios	.65	0.236	.029	$-.039$	$-.318$
Qwen-bios	.85	0.948	.037	$-.016$	$-.405$
Qwen-bios	.97	1.703	.094	$-.023$	$-.413$
Qwen-civil	.65	0.297	.041	$-.059$	$-.428$
Qwen-civil	.85	0.592	.057	$-.023$	$-.233$
Qwen-civil	.97	1.224	.084	$+.007$	$+.097$

Table 5: Within-stratum slopes (b, raw accuracy points per Δ_{cue} unit) and correlations (r), 15 seeds per stratum. SD(w) is the within-stratum SD of worst-group accuracy.

specification	β	p
z within cell + stratum FE (mismatched)	$-.0254$	.147
z within stratum + stratum FE	$-.0124$	.060
raw Δ_{cue} + stratum FE	$-.0129$	.084
both z within stratum	$-.2166$	.015

Table 6: The same within-recipe effect under four combinations of predictor scaling and fixed effects (n=135). The first row is the most conservative and is the interval used for the equivalence bound.

D Repair Detail

This section supports §6 with the overall-accuracy cost of each repair and the budgeted allocation experiment.

Intervention layers. LEACE and the fixed-depth ablation act at layer $\text{round}(20\, n_{\text{layers}}/24)$, which is layer 20 in the 24-layer models and the same relative depth otherwise. The LEACE eraser and the ablation directions are fit on training-split texts only. The prospective arm instead selects, from training-side texts alone, the layer whose direction projection most reduces training-side Δ_{cue}.

D.1 Overall-Accuracy Cost

Table 8 reports mean post-repair overall accuracy next to the worst-group gains of Table 1. Post-LEACE overall accuracy is within 0.03 of DFR's in every cell. The two cells where hard ablation

margin	% of ρ effect	TOST p	verdict
.020	5.2	.622	not equivalent
.030	7.7	.396	not equivalent
.040	10.3	.202	not equivalent
.050	12.9	.080	not equivalent
.054	14.0	.050	boundary
.060	15.5	.025	equivalent
.075	19.4	.003	equivalent
.100	25.8	.000	equivalent

Table 7: TOST margin sensitivity for the within-recipe effect. "Equivalent" means effects larger than the margin are rejected at α=.05.

cell	ablate	LEACE	DFR	AFR
Qwen-civil	.771	.762	.782	.755
Pythia-civil	.627	.772	.790	.775
OLMo-civil	.686	.790	.807	.785
Qwen-bios	.938	.937	.948	.931
Pythia-bios	.926	.934	.949	.927
OLMo-bios	.949	.946	.957	.940

Table 8: Mean post-repair overall accuracy (n=18/cell). Post-repair overall accuracy was not logged for the prospectively-selected-layer arm.

is strongly harmful for worst-group accuracy also show large overall-accuracy losses.

D.2 Allocation Under a Budget

Each high-seed cell (45 models) is evaluated independently, with each policy's ranking rule fixed in advance, so nothing is fitted. Policies rank by descending Δ_{cue} (within-cell standardized), by headroom (1−baseline worst, identical to ranking by visible failure), by assigned ρ (stratum-mean expectation, since ρ cannot rank within a stratum), or by realized gain (oracle). Repairing the top 15: random $+0.091$, Δ_{cue} $+0.200$, headroom $+0.201$, highest-ρ $+0.201$, oracle $+0.205$ (per-cell Δ_{cue} selections $+0.303/+0.116/+0.182$ against oracle $+0.307/+0.118/+0.189$). At this budget, selection is chiefly correlation-level identification, consistent with the fragile fixed-recipe effect (§7).

Table 9 reports the full budget sweep. At budgets below one stratum the measurement ranks within the high-ρ stratum, which ρ cannot, and the Δ_{cue} selection is far outside what random selection attains. Across the three cells, the probability that a random subset matches it is at most 0.014 at budget 5 and below 10^{-4} at budget 10 (10,000 draws per cell). Headroom itself correlates with gain at r=0.79, the confound reported in §6.

budget	random	Δ_{cue}	headroom	ρ	oracle
5	.091	.224	.218	.201	.281
10	.091	.214	.210	.201	.237
15	.091	.200	.201	.201	.205
20	.091	.173	.176	.170	.181
30	.091	.138	.139	.138	.142

Table 9: Mean realized LEACE gain of the selected models by repair budget (of 45 per cell, averaged over the three cells). The ρ policy uses stratum-mean expectations, since ρ cannot rank within a stratum.

	labeled, k/group				Δ_{cue}, m unlabeled		
cell	5	10	25	50	25	100	300
Qwen-civil	.70	.83	.92	.96	.87	.90	.91
Pythia-civil	.68	.82	.92	.96	.72	.76	.77
OLMo-civil	.63	.78	.90	.95	.84	.89	.90
Qwen-bios	.34	.51	.72	.84	.92	.93	.93
Pythia-bios	.58	.75	.87	.94	.81	.81	.81
OLMo-bios	.37	.53	.73	.86	.87	.88	.88

Table 10: Correlation with true worst-group accuracy per cell (n=18 models each).

E Additional Analyses

This section collects the secondary analyses referenced from the main text.

E.1 Label-Efficiency Detail

Table 10 gives the correlation between each estimator and true worst-group accuracy across the 18 models of each cell. It compares direct estimation from k group-labeled examples per group, which also requires the roughly 4,000 labels behind the group split, against Δ_{cue} from m unlabeled texts that contain the cue.

E.2 Early Measurability

Δ_{cue} measured after one of four training epochs already correlates -0.886 with final worst-group accuracy, and the epoch-1 signal keeps 108.4% of the final measurement's R^2 ($0.785/0.724$). It is not, however, a useful early-warning tool. Epoch-1 overall accuracy predicts final worst-group accuracy equally well (r=+0.875). After cell fixed effects and exact ρ control, no epoch-1 signal, including Δ_{cue}, adds predictive value (ΔR^2=0.000, p=0.65), because within a stratum the epoch-1 signals share their variance. Table 11 lists all predictors.

E.3 The Unused Instrument

Because ρ is randomized, it is formally available as an instrument for Δ_{cue}. On the high-seed sam-

epoch-1 predictor	r	ctrl. β	ctrl. p
Δ_{cue}	$-.886$	$+.013$	.669
final Δ_{cue} (ceiling)	$-.851$	$-.012$	.327
worst-group (oracle)	$+.836$	$-.049$	.621
worst-class (label-free)	$+.710$	$-.048$	.735
overall accuracy	$+.875$	$+.019$	.950
train loss	$+.811$	$+.115$	.742

Table 11: Epoch-1 predictors of final worst-group accuracy (n=72, Qwen-civil and Pythia-civil). Raw correlations are uniformly strong. Under cell fixed effects and exact ρ control, none survives.

model	worst	overall	Δ_{cue}
s-nlp/roberta_toxicity	.838	.894	$+2.28$
martin-ha/toxic-comment	.807	.854	$+0.86$
unitary/toxic-bert	.790	.846	$+0.81$
JungleLee/bert-toxic	.735	.828	$+1.04$
cardiffnlp/twitter-offensive	.662	.799	$+0.12$
CNERG/dehatebert-mono-en	.580	.667	$+0.26$
citizenlab/distilbert-toxicity	.565	.726	-0.04
facebook/dynabench-r4	.330	.626	$+1.86$

Table 12: Public classifiers under the study protocol, sorted by worst-group accuracy. Capability ordering and worst-group ordering nearly coincide.

ple the first stage is strong by construction ($+15.0$ Δ_{cue} units per unit ρ), and two-stage least squares gives -0.077 against an OLS estimate of -0.062 (ratio 1.24, cell fixed effects). We do not interpret this coefficient causally because the exclusion restriction, that ρ affects worst-group accuracy only through Δ_{cue}, is likely violated. We therefore report it only as a descriptive sensitivity analysis.

E.4 The Public-Classifier Arm

Table 12 evaluates eight off-the-shelf toxicity classifiers under the same protocol as the main study (same held-out split, same edit, train-side threshold calibration, toxic-logit scoring for multi-label heads). Worst-group accuracy tracks overall accuracy almost perfectly (Spearman $+0.98$), and Δ_{cue} does not predict worst-group accuracy (r=$+0.06$, p=0.88). This is the scope limit reported in Limitations. In this small heterogeneous set, capability differences dominate, and Δ_{cue} does not separate from them.